

Qatar-4b

R-2494

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_211201 · created 2026-08-24T01:37:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459549.6604 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.12 +/- 0.042
Transit depth (Rp/Rs)^2	1.45 +/- 1.0 [%]
Orbital Inclination (inc)	85.06 +/- 1.99
Airmass coefficient 1 (a1)	1.408 +/- 0.038
Airmass coefficient 2 (a2)	-0.31 +/- 0.1
Scatter in the residuals of the lightcurve fit is	2.73 %
Best Comparison Star	#1 - [88, 100]
Optimal Method	PSF photometry
Transit Duration (day)	0.071 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.12 $\pm$ 0.042 vs 0.138 (deviation 0.43 σ)
Duration fit vs published	0.071 d vs 0.0892 d (deviation 1.01 σ)
Mid-time O-C	6.9 min
Depth SNR	2.9
Residual scatter	2.73 %

Light curve

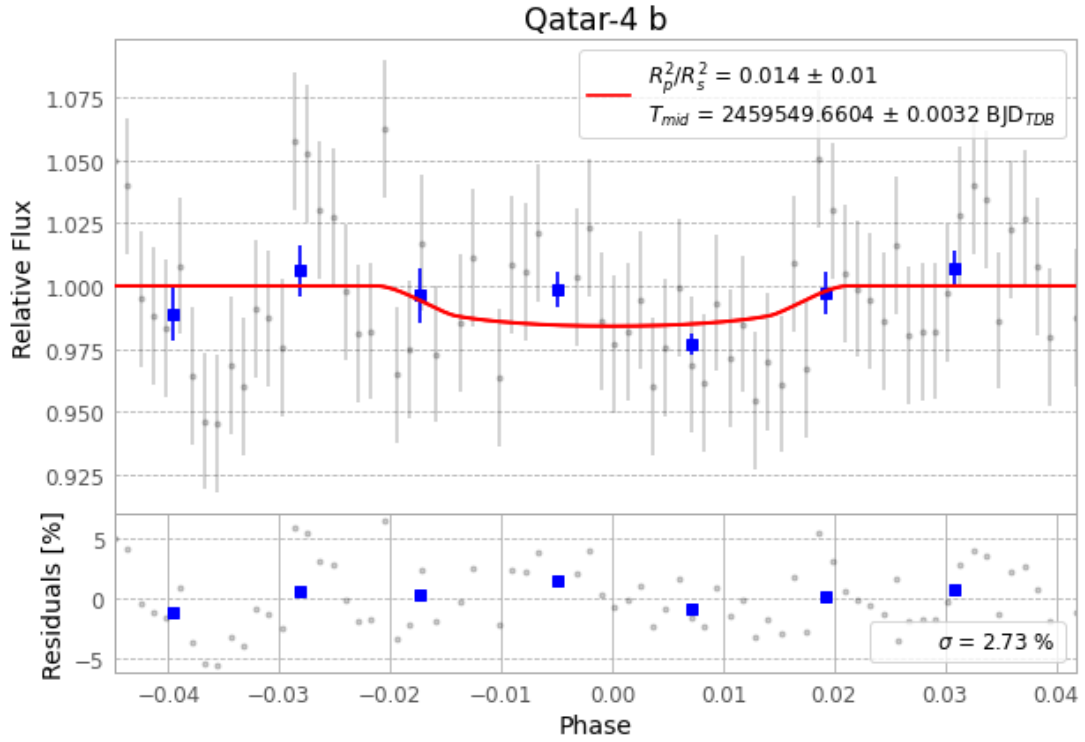


Figure 1 — FinalLightCurve_Qatar-4 b_2021-11-30.png (SHA-256 f7f00e523bba24fd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53d0ecb0146e2f27...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-4211201014812.FITS → Qatar-4211201053310.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-211201.log (9c7d077f3c82...), FinalLightCurve_Qatar-4 b_2021-11-30.png (f7f00e523bba...), FinalLightCurve_Qatar-4 b_2021-11-30.csv (1513631a586c...)

Compute resources

Wall time 164.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)